

The Gaussian Correlation Inequality

After T. Royen (2014) and R. Latała & D. Matlak (2015)

1 Introduction

Theorem 1.1 (Gaussian correlation inequality). For any closed symmetric convex sets $K, L \subseteq \mathbb{R}^d$ and any centered Gaussian measure μ on $\mathbb{R}^d$,

$$\mu(K \cap L) \geq \mu(K) \mu(L). \quad (1)$$

These notes prove Theorem 1.1 by Royen's analytic argument, following Latała–Matlak's Gaussian-only exposition. The inputs are the Laplace transform of a squared Gaussian, a determinant expansion, dominated convergence, and integration by parts. Historical background and further remarks are collected in Section 7.

Notation 1.2.

- $\mathcal{N}(0, C)$: centered Gaussian distribution with covariance C .
- $M_{n \times m}$: real $n \times m$ matrices. $A \succ 0$ (resp. $A \succcurlyeq 0$) means A is symmetric positive (semi)definite.
- For $A = (a_{ij})_{i,j \leq n}$ and $J \subseteq [n] := \{1, \dots, n\}$, $A_J := (a_{ij})_{i,j \in J}$ is the principal submatrix, $|J|$ the cardinality, and $|A|$ the determinant, defined by the Leibniz (permutation) formula

$$|A| = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i\sigma(i)}, \quad \sigma \text{ ranging over the permutations of } [n]. \quad (2)$$

We use freely its standard consequences: $|AB| = |A| |B|$, invariance under adding a multiple of one row to another, and $\begin{vmatrix} B & C \\ 0 & D \end{vmatrix} = |B| |D|$ for square diagonal blocks.

- I_n : $n \times n$ identity.
- For $\varepsilon \in \{-1, 1\}^n$, $x \in (0, \infty)^n$: $\sqrt{x}_\varepsilon := (\varepsilon_i \sqrt{x_i})_{i \leq n}$.
- A few letters are reused in different sections, always with a local definition: μ is a centered Gaussian measure, but also the scale parameter of Theorem 3.6 and the eigenvalues $\mu_{J_1, J_2}(i)$ in Sections 4 and 7.4; K is a symmetric convex set, but also the box $[0, s_1] \times \dots \times [0, s_n]$ in Section 4 and compact sets K_m ; λ is a Laplace variable, but also an eigenvalue in Section 7.2; g is a noncentral gamma density g_α , but also the dominating function in Theorem 3.3 and the functions g_i of Theorem 6.10.

We will use three standard facts without further comment:

Lemma 1.3 (Square root). For any $A \succcurlyeq 0$ there exists a unique $A^{1/2} \succcurlyeq 0$ with $(A^{1/2})^2 = A$, and $A^{1/2}$ is invertible iff $A \succ 0$.

(Spectral theorem: diagonalize $A = O\Lambda O^\top$ with $\Lambda \succcurlyeq 0$, set $A^{1/2} := O\Lambda^{1/2}O^\top$; uniqueness is standard.)

Lemma 1.4 (Sylvester determinant identity). For $A_1 \in M_{n \times m}$ and $A_2 \in M_{m \times n}$, $|I_n + A_1 A_2| = |I_m + A_2 A_1|$.

Proof. Compute $D := \begin{vmatrix} I_n & -A_1 \\ A_2 & I_m \end{vmatrix}$ two ways. Row-reduce $R_1 \leftarrow R_1 + A_1 R_2$ to clear the $(1, 2)$ block: the matrix becomes $\begin{pmatrix} I_n + A_1 A_2 & 0 \\ A_2 & I_m \end{pmatrix}$, so $D = |I_n + A_1 A_2|$. Alternatively, row-reduce $R_2 \leftarrow R_2 - A_2 R_1$: the matrix becomes $\begin{pmatrix} I_n & -A_1 \\ 0 & I_m + A_2 A_1 \end{pmatrix}$, so $D = |I_m + A_2 A_1|$. $\square$

Lemma 1.5 (Strict separation; Hahn–Banach). Let $K_1, K_2 \subseteq \mathbb{R}^d$ be disjoint closed convex sets with K_1 bounded. There exist $w \in \mathbb{R}^d \setminus \{0\}$ and $\alpha_1 < \alpha_2$ in $\mathbb{R}$ with $\langle w, x \rangle \leq \alpha_1$ for $x \in K_1$ and $\langle w, x \rangle \geq \alpha_2$ for $x \in K_2$.

(Standard corollary of Hahn–Banach in finite dimension; see e.g. Rockafellar, *Convex Analysis*, Cor. 11.4.2.)

Besides these three, the following standard facts of analysis and probability are also imported without proof: a probability law on $\mathbb{R}^n$ is determined by its characteristic function, and Fourier inversion for integrable characteristic functions (used for the bivariate normal); a linear image VX of a centered Gaussian vector X is centered Gaussian with covariance $VCV^\top$, and for $C \succ 0$ the law $\mathcal{N}(0, C)$ has density ϕ_C (change of variables); the eigenvalues of a symmetric matrix depend continuously on its entries; the Stone–Weierstrass theorem, Urysohn’s lemma and the regularity of finite Borel measures on $\mathbb{R}^n$; the boundary of a convex subset of $\mathbb{R}^n$ has Lebesgue measure zero; and the strict log-convexity of Γ on $(0, \infty)$, together with the numerical value of its minimum. (The uniqueness part of Theorem 1.3 is never used.)

2 Setup and reduction to rectangles

Definition 2.1 (Centered Gaussian measure). For $C \succ 0$, the measure $\mathcal{N}(0, C)$ on $\mathbb{R}^n$ has density

$$\phi_C(x) = (2\pi)^{-n/2} |C|^{-1/2} \exp\left(-\frac{1}{2} \langle C^{-1}x, x \rangle\right).$$

More generally, a random vector X in $\mathbb{R}^n$ is *centered Gaussian with covariance* $C \succ 0$ if $X = AZ$ for some $A \in M_{n \times m}$ with $AA^\top = C$ and $Z \sim \mathcal{N}(0, I_m)$. Its law depends only on C (its characteristic function is $u \mapsto e^{-\langle Cu, u \rangle/2}$); we denote it $\mathcal{N}(0, C)$, and for $C \succ 0$ it is the measure with density ϕ_C above. A centered Gaussian measure is a measure $\mathcal{N}(0, C)$ with $C \succ 0$.

In particular, if $G = AZ \sim \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^d$ and $V \in M_{n \times d}$, then $VG = (VA)Z$ is centered Gaussian with covariance $V\Sigma V^\top$, which may be singular even when $\Sigma \succ 0$ (e.g. whenever $n > d$). We assume $C \succ 0$ throughout the proof of Theorem 2.4; singular $C \succ 0$ is reduced to it by Theorem 2.5 below.

Definition 2.2. $K \subseteq \mathbb{R}^n$ is *symmetric* if $K = -K$. A *symmetric slab* is $S_{v,t} := \{x \in \mathbb{R}^n : |\langle v, x \rangle| \leq t\}$ for $v \in \mathbb{R}^n \setminus \{0\}$, $t > 0$.

Lemma 2.3 (Slab approximation). Every nonempty *bounded* closed symmetric convex set $K \subseteq \mathbb{R}^d$ is a countable intersection of symmetric slabs $S_{v,t}$.

Proof. Fix $R < \infty$ with $K \subseteq B(0, R)$, and let $\mathcal{V} := \{(v, t) \in (\mathbb{Q}^d \setminus \{0\}) \times \mathbb{Q}_{>0} : K \subseteq S_{v,t}\}$, a countable set; trivially $K \subseteq \bigcap_{(v,t) \in \mathcal{V}} S_{v,t}$. For the reverse inclusion we show that each $y \notin K$ is excluded by some slab in $\mathcal{V}$.

Fix $y \notin K$ and pick $\varepsilon > 0$ with $B(y, \varepsilon) \cap K = \emptyset$. The sets $\overline{B(y, \varepsilon/2)}$ (bounded) and K (closed convex) are disjoint, so Theorem 1.5 yields a separating functional; replacing it by its negative if necessary, we obtain $w \in \mathbb{R}^d \setminus \{0\}$ and $\alpha_1 < \alpha_2$ with $\langle w, x \rangle \leq \alpha_1$ for $x \in K$ and $\langle w, x \rangle \geq \alpha_2$ for $x \in \overline{B(y, \varepsilon/2)}$. Set $a := \sup_{x \in K} \langle w, x \rangle \leq \alpha_1$. Since K is nonempty, symmetric, and convex, $0 = \frac{1}{2}x + \frac{1}{2}(-x) \in K$ for any $x \in K$, so $a \geq \langle w, 0 \rangle = 0$; and $K = -K$ gives $\sup_{x \in K} |\langle w, x \rangle| \leq a$. The minimum of $\langle w, \cdot \rangle$ over $\overline{B(y, \varepsilon/2)}$ is attained at $y - \frac{\varepsilon}{2}w/|w|$, so

$$\langle w, y \rangle = \min_{z \in \overline{B(y, \varepsilon/2)}} \langle w, z \rangle + \frac{\varepsilon}{2}|w| \geq \alpha_2 + \frac{\varepsilon}{2}|w| \geq a + \frac{\varepsilon}{2}|w| > 0,$$

a strictly positive margin over a .

Now fix δ with $0 < \delta < \frac{\varepsilon|w|}{2(|y| + R + 1)}$ and choose $w' \in \mathbb{Q}^d$ with $|w' - w| < \delta$. Then $w' \neq 0$: since $0 \in K$ and $B(y, \varepsilon) \cap K = \emptyset$ we have $\varepsilon \leq |y|$, so $\delta < |w|/2$ and $|w'| > |w|/2 > 0$; thus $S_{w',a'}$ below is a slab. Using $|x| \leq R$ on K , $\sup_{x \in K} |\langle w, x \rangle| \leq a$, and $\langle w, y \rangle \geq a + \frac{\varepsilon}{2}|w|$,

$$\sup_{x \in K} |\langle w', x \rangle| \leq a + \delta R, \quad |\langle w', y \rangle| \geq a + \frac{\varepsilon}{2}|w| - \delta|y|.$$

Taking a rational $a' \in [a + \delta R, a + \delta R + \delta/2]$ (an interval of positive length, so a' exists, and $a' > 0$), the choice of δ gives $\delta(|y| + R + \frac{1}{2}) < \frac{\varepsilon}{2}|w|$, which gives both $K \subseteq S_{w',a'}$ and $y \notin S_{w',a'}$, as required. $\square$

We now reduce Theorem 1.1 to a statement about slabs. If K or L is empty both sides of (1) vanish, so assume both are nonempty. We proceed in two steps, each by continuity of the finite measure μ (from below for the increasing truncations of the first step, from above for the decreasing slab-intersections of the second). First, putting $K_m := K \cap [-m, m]^d$ and $L_m := L \cap [-m, m]^d$ (nonempty for every m since $0 \in K \cap L$, bounded closed symmetric convex, increasing to K, L), we get $\mu(K_m) \rightarrow \mu(K)$, $\mu(L_m) \rightarrow \mu(L)$, and $\mu(K_m \cap L_m) \rightarrow \mu(K \cap L)$, so it suffices to treat *bounded* K, L . Second, Theorem 2.3 writes each bounded set as a countable intersection of symmetric slabs; intersecting the first N of them and letting $N \rightarrow \infty$, the same continuity reduces the GCI to the case where K and L are *finite* intersections of symmetric slabs. Writing

$$K = \{x : |\langle v_i, x \rangle| \leq t_i, 1 \leq i \leq n_1\}, \quad L = \{x : |\langle v_i, x \rangle| \leq t_i, n_1+1 \leq i \leq n\},$$

setting $n = n_1 + n_2$ and $X_i := \langle v_i, G \rangle$ for $G \sim \mu$, the vector $X = (X_1, \dots, X_n)$ is centered Gaussian with a covariance $C \succ 0$ that is in general singular, and the GCI reduces to the following theorem together with Theorem 2.5:

Theorem 2.4 (GCI, analytic form). Let $X = (X_1, \dots, X_n)$ be centered Gaussian with $C \succ 0$, $n = n_1 + n_2$. For any $t_1, \dots, t_n > 0$,

$$\mathbb{P}(|X_i| \leq t_i, i \leq n) \geq \mathbb{P}(|X_i| \leq t_i, i \leq n_1) \mathbb{P}(|X_i| \leq t_i, n_1+1 \leq i \leq n). \quad (3)$$

Lemma 2.5 (Singular covariance). If (3) holds for every centered Gaussian X with covariance $C \succ 0$, then it holds for every centered Gaussian X with covariance $C \succcurlyeq 0$ (and the same $t_i > 0$).

Proof. Let $X = AZ$ have covariance $C \succcurlyeq 0$, let $Y \sim \mathcal{N}(0, I_n)$ be independent of Z , and for $\varepsilon > 0$ put $X^\varepsilon := X + \varepsilon Y = \begin{bmatrix} A & \varepsilon I_n \end{bmatrix} \begin{pmatrix} Z \\ Y \end{pmatrix}$. Then X^ε is centered Gaussian with covariance $C + \varepsilon^2 I_n \succ 0$, so (3) holds for X^ε . For each i , $\mathbb{P}(|X_i| = t_i) = 0$: if $C_{ii} > 0$ then $X_i \sim \mathcal{N}(0, C_{ii})$ has a density, and if $C_{ii} = 0$ then $\mathbb{E}X_i^2 = 0$, so $X_i = 0 < t_i$ a.s. Since $X_i^\varepsilon \rightarrow X_i$ pointwise as $\varepsilon \rightarrow 0$, on the full-probability event $\bigcap_i \{|X_i| \neq t_i\}$ we get $\mathbf{1}\{|X_i^\varepsilon| \leq t_i\} \rightarrow \mathbf{1}\{|X_i| \leq t_i\}$ for every i , hence also for the products of these indicators over $i \leq n$, $i \leq n_1$, and $n_1 < i \leq n$. By dominated convergence (all bounded by 1), each of the three probabilities in (3) for X^ε converges to the corresponding probability for X , and the inequality passes to the limit. $\square$

3 Auxiliary lemmas

3.1 Laplace transform of squared Gaussian vectors

Lemma 3.1. For $X \sim \mathcal{N}(0, C)$ in $\mathbb{R}^n$ and $\lambda_i \geq 0$,

$$\mathbb{E} \exp\left(-\sum_{i=1}^n \lambda_i X_i^2\right) = |I_n + 2\Lambda C|^{-1/2}, \quad \Lambda := \text{diag}(\lambda_1, \dots, \lambda_n).$$

Proof.

1. Standard normal quadratic. For $Y \sim \mathcal{N}(0, I_n)$ and a symmetric matrix B with $I_n - 2B \succ 0$, combine the density of Y with the factor $e^{\langle BY, Y \rangle}$: the exponent is $-\frac{1}{2}\langle y, y \rangle + \langle By, y \rangle = -\frac{1}{2}\langle (I_n - 2B)y, y \rangle$, so

$$\mathbb{E} e^{\langle BY, Y \rangle} = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle (I_n - 2B)y, y \rangle} dy = |I_n - 2B|^{-1/2}, \quad (4)$$

using $\int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Ay, y \rangle} dy = (2\pi)^{n/2} |A|^{-1/2}$ (change variables $y = A^{-1/2}z$; $A = I_n - 2B$).

2. Reduction to the standard case. Set $A := C^{1/2}$ (Theorem 1.3); then $C = AA^\top$ and $X := AY \sim \mathcal{N}(0, C)$. Taking $B := -A^\top \Lambda A$ in (4): since $\lambda_i \geq 0$ gives $\Lambda \succcurlyeq 0$, hence $B \preccurlyeq 0$ and $I_n - 2B \succcurlyeq I_n \succ 0$. The exponent becomes $\langle BY, Y \rangle = -\langle \Lambda AY, AY \rangle = -\sum_i \lambda_i (AY)_i^2 = -\sum_i \lambda_i X_i^2$, so

$$\mathbb{E} \exp\left(-\sum_i \lambda_i X_i^2\right) = |I_n + 2A^\top \Lambda A|^{-1/2}.$$

3. Commuting the factors. Apply Theorem 1.4 with $A_1 := A^\top$, $A_2 := 2\Lambda A$: $|I_n + 2A^\top \Lambda A| = |I_n + 2\Lambda A A^\top| = |I_n + 2\Lambda C|$.

□

3.2 A determinant identity

Lemma 3.2.

- (i) For $A \in M_{n \times n}$,

$$|I_n + A| = 1 + \sum_{\emptyset \neq J \subseteq [n]} |A_J|. \quad (5)$$

- (ii) For $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \in M_{n \times n}$ with blocks $A_{ij} \in M_{n_i \times n_j}$, $n_1 + n_2 = n$, and $A_{11}, A_{22} \succ 0$,

$$|A| = |A_{11}| |A_{22}| \cdot |I_{n_1} - A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}|. \quad (6)$$

If moreover $A \succ 0$, then $0 \preccurlyeq A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} \preccurlyeq I_{n_1}$.

Proof.

- (i) By the Leibniz formula (2), $|I_n + A| = \sum_{\sigma} \text{sgn}(\sigma) \prod_i (\delta_{i\sigma(i)} + a_{i\sigma(i)})$. Expanding the product, each summand corresponds to a choice of $J \subseteq [n]$ (positions where $a_{i\sigma(i)}$ is selected); non-zero contributions require $\sigma(i) = i$ for $i \notin J$. The σ 's fixing $[n] \setminus J$ correspond to permutations of J , giving $\sum_J |A_J|$; the $J = \emptyset$ term is 1.
- (ii) Let $A_{11}^{1/2}, A_{22}^{1/2}$ be the positive symmetric square roots (Theorem 1.3), invertible since $A_{11}, A_{22} \succ 0$. Setting $P := A_{11}^{-1/2} A_{12} A_{22}^{-1/2}$ and $Q := A_{22}^{-1/2} A_{21} A_{11}^{-1/2}$, a direct block multiplication gives

$$A = \text{diag}(A_{11}^{1/2}, A_{22}^{1/2}) \begin{pmatrix} I & P \\ Q & I \end{pmatrix} \text{diag}(A_{11}^{1/2}, A_{22}^{1/2}),$$

so $|A| = |A_{11}| |A_{22}| \cdot \left| \begin{pmatrix} I & P \\ Q & I \end{pmatrix} \right|$. Row-reduce by $R_1 \leftarrow R_1 - P R_2$ (determinant invariant):

$$\begin{pmatrix} I & P \\ Q & I \end{pmatrix} \rightarrow \begin{pmatrix} I - PQ & 0 \\ Q & I \end{pmatrix},$$

block-triangular with determinant $|I - PQ|$. Substituting $PQ = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}$ gives (6).

When $A \succ 0$: for all $s, t \in \mathbb{R}$ and $x \in \mathbb{R}^{n_1}, y \in \mathbb{R}^{n_2}$,

$$\langle A(sx, ty)^\top, (sx, ty)^\top \rangle = s^2 \langle A_{11}x, x \rangle + 2st \langle A_{21}x, y \rangle + t^2 \langle A_{22}y, y \rangle \geq 0.$$

Nonnegativity of this quadratic in (s, t) forces $\langle A_{21}x, y \rangle^2 \leq \langle A_{11}x, x \rangle \langle A_{22}y, y \rangle$. Setting $x := A_{11}^{-1/2}u$, $y := A_{22}^{-1/2}v$: $\langle Qu, v \rangle^2 \leq |u|^2 |v|^2$, i.e., $\|Q\|_{\text{op}} \leq 1$, equivalently $0 \preccurlyeq Q^\top Q \preccurlyeq I_{n_1}$. For A symmetric, $Q = P^\top$, so $A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2} = PP^\top = Q^\top Q$.

□

3.3 Density of $Z(\tau)$ and the interchange lemma

Partition

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad C_{ij} \in M_{n_i \times n_j}, \quad n_1 + n_2 = n, \quad (7)$$

and define the interpolating family

$$C(\tau) := \begin{pmatrix} C_{11} & \tau C_{12} \\ \tau C_{21} & C_{22} \end{pmatrix} = \tau C(1) + (1 - \tau) C(0), \quad \tau \in [0, 1]. \quad (8)$$

Since $C(0) = \text{diag}(C_{11}, C_{22}) \succ 0$ and $C(1) = C \succ 0$, each $C(\tau) \succ 0$ by convexity. Let $X(\tau) \sim \mathcal{N}(0, C(\tau))$ and $Z_i(\tau) := \frac{1}{2} X_i(\tau)^2$.

Lemma 3.3 (Density and interchange). Let $f(x, \tau)$ denote the density of $Z(\tau)$ on $(0, \infty)^n$.

(i)

$$f(x, \tau) = |C(\tau)|^{-1/2} (4\pi)^{-n/2} \frac{1}{\sqrt{x_1 \cdots x_n}} \sum_{\varepsilon \in \{-1, 1\}^n} e^{-\langle C(\tau)^{-1} \sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle} \mathbf{1}_{(0, \infty)^n}(x). \quad (9)$$

(ii) For any Borel $K \subseteq [0, \infty)^n$ and $\lambda_i \geq 0$,

$$\int_K e^{-\sum_i \lambda_i x_i} \frac{\partial}{\partial \tau} f(x, \tau) dx = \frac{\partial}{\partial \tau} \int_K e^{-\sum_i \lambda_i x_i} f(x, \tau) dx. \quad (10)$$

Proof.

(i) The density of $X(\tau)$ is $\phi_{C(\tau)}$ from Theorem 2.1. On each sign orthant $\mathbb{R}_\varepsilon^n := \{y : \text{sgn}(y_i) = \varepsilon_i\}$, the map $y \mapsto x_i := \frac{1}{2}y_i^2$ is a diffeomorphism onto $(0, \infty)^n$ with inverse $y_i = \varepsilon_i \sqrt{2x_i}$ and Jacobian $dy_i/dx_i = \varepsilon_i/\sqrt{2x_i}$; absolute Jacobian $\prod_i (2x_i)^{-1/2}$. Summing over the 2^n sign patterns and using $\langle C(\tau)^{-1}y, y \rangle = 2\langle C(\tau)^{-1}\sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle$ (by $y_i = \varepsilon_i \sqrt{2x_i}$):

$$\begin{aligned} f(x, \tau) &= \sum_{\varepsilon} \phi_{C(\tau)}(\sqrt{2x_\varepsilon}) \cdot \prod_i (2x_i)^{-1/2} \\ &= |C(\tau)|^{-1/2} (2\pi)^{-n/2} 2^{-n/2} (x_1 \cdots x_n)^{-1/2} \sum_{\varepsilon} e^{-\langle C(\tau)^{-1} \sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle}, \end{aligned}$$

which is (9) after combining $(2\pi)^{-n/2} 2^{-n/2} = (4\pi)^{-n/2}$.

(ii) We build a τ -uniform integrable dominating function for $\partial_\tau f$ via four steps.

- (a) Eigenvalue lower bound. Each $C(\tau) \succ 0$, and $\tau \mapsto \lambda_{\min}(C(\tau)^{-1}) = 1/\lambda_{\max}(C(\tau))$ is continuous and positive on compact $[0, 1]$, so bounded below by some $a > 0$. Then $\langle C(\tau)^{-1} \sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle \geq a |\sqrt{x_\varepsilon}|^2 = a \sum_i x_i$.
- (b) Derivative formula. Differentiating $C(\tau)C(\tau)^{-1} = I_n$, $\frac{d}{d\tau} C(\tau)^{-1} = -C(\tau)^{-1} C'(\tau) C(\tau)^{-1}$, with $C'(\tau) = C(1) - C(0)$ constant. Since $\tau \mapsto C(\tau)^{-1}$ is continuous on compact $[0, 1]$, $\|\frac{d}{d\tau} C(\tau)^{-1}\|_{\text{op}}$ is bounded, say by $b < \infty$. Since $|C(\tau)|$ is a polynomial in τ that is positive on $[0, 1]$, the map $\tau \mapsto |C(\tau)|^{-1/2}$ is C^1 on $[0, 1]$, so it and its derivative are bounded there, say by $M, M' < \infty$. Hence $|C(\tau)|^{-1/2} \leq M$, $|\partial_\tau |C(\tau)|^{-1/2}| \leq M'$, and $|\langle \frac{d}{d\tau} C(\tau)^{-1} \sqrt{x_\varepsilon}, \sqrt{x_\varepsilon} \rangle| \leq b |\sqrt{x_\varepsilon}|^2 = b \sum_i x_i$.
- (c) Dominating function. Differentiating (9) in τ (product rule on the prefactor $|C(\tau)|^{-1/2}$ and the exponent), each of the 2^n sign-orthant summands is bounded by

$$(M' + Mb \sum_i x_i) (4\pi)^{-n/2} (x_1 \cdots x_n)^{-1/2} e^{-a \sum_i x_i}.$$

Summing over ε and absorbing constants,

$$\sup_{\tau \in [0, 1]} |\partial_\tau f(x, \tau)| \leq g(x) := c \frac{1 + \sum_i x_i}{\sqrt{x_1 \cdots x_n}} e^{-a \sum_i x_i}$$

for some $c = c(C, n)$. Since $1 + \sum_i x_i \leq \prod_i (1 + x_i)$, we have $g(x) \leq c \prod_i x_i^{-1/2} (1 + x_i) e^{-ax_i}$, which factorises; as $\int_0^\infty x^{-1/2} (1 + x) e^{-ax} dx < \infty$, Tonelli gives $\int g < \infty$.

- (d) Interchange. For any Borel $K \subseteq [0, \infty)^n$ and $\lambda_i \geq 0$, $e^{-\sum_i \lambda_i x_i} |\partial_\tau f| \leq g$ uniformly in τ . Dominated convergence applied to the difference quotient $(f(x, \tau + h) - f(x, \tau))/h \rightarrow \partial_\tau f(x, \tau)$ gives (10).

□

3.4 The noncentral gamma and the h -family

Definition 3.4. For $\alpha > 0$, $y \geq 0$, the *noncentral gamma density* is

$$g_\alpha(x, y) := e^{-x-y} \sum_{k=0}^{\infty} \frac{x^{k+\alpha-1}}{\Gamma(k+\alpha)} \frac{y^k}{k!}, \quad x > 0.$$

Remark 3.5. Equivalently,

$$g_\alpha(x, y) = \sum_{k \geq 0} e^{-y} \frac{y^k}{k!} g_{\alpha+k}(x, 0),$$

a Poisson(y) mixture of the central gamma($\alpha + k$) densities $g_{\alpha+k}(x, 0) = x^{\alpha+k-1} e^{-x} / \Gamma(\alpha + k)$; so $g_\alpha(\cdot, y)$ is a probability density for each $y \geq 0$.

Definition 3.6. For a scale parameter $\mu > 0$ (unrelated to the Gaussian measure of Theorem 1.1), $\alpha \in (0, \infty)^n$, and $Y = (Y_1, \dots, Y_n)$ with $Y_i \geq 0$ a.s.,

$$h_{\alpha, \mu, Y}(x) := \mathbb{E} \prod_{i=1}^n \frac{1}{\mu} g_{\alpha_i} \left(\frac{x_i}{\mu}, Y_i \right), \quad x \in (0, \infty)^n.$$

Lemma 3.7 (Properties of the h -family). Write $h_\alpha := h_{\alpha, \mu, Y}$.

- (i) $h_\alpha \geq 0$ and $\int h_\alpha dx = 1$.
- (ii) If $\alpha_i > 1$, then $h_\alpha(x) \rightarrow 0$ as $x_i \rightarrow 0+$, and

$$\frac{\partial}{\partial x_i} h_\alpha = \frac{1}{\mu} (h_{\alpha - e_i} - h_\alpha). \quad (11)$$

- (iii) If $\alpha \in (1, \infty)^n$, then for any $J \subseteq [n]$ and $\lambda_i \geq 0$, $\partial^{|J|} h_\alpha / \partial x_J \in L_1((0, \infty)^n)$ and

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\alpha dx = \left(\prod_{i \in J} \lambda_i \right) \int e^{-\sum_i \lambda_i x_i} h_\alpha dx. \quad (12)$$

Proof.

- (i) $\int_0^\infty \mu^{-1} g_\alpha(x_i/\mu, Y_i) dx_i = \int_0^\infty g_\alpha(u, Y_i) du = 1$ for each ω (Theorem 3.5); Fubini gives $\int h_\alpha = \mathbb{E}[1] = 1$.
- (ii) We split into two parts.

- (a) Gamma bound. $\log \Gamma$ is strictly convex on $(0, \infty)$, and since $\Gamma(1) = \Gamma(2) = 1$, Γ has a unique minimum at some $s_0 \in (1, 2)$ with $\Gamma(s_0) > \frac{1}{2}$ (numerically $s_0 \approx 1.462$, $\Gamma(s_0) \approx 0.886$). Hence

$$\Gamma(s) \geq \frac{1}{2} \quad \text{for all } s > 0, \quad (13)$$

giving $\Gamma(\alpha)^{-1} \leq 2$ for the $k = 0$ term. For $k \geq 1$, iterating $\Gamma(s+1) = s\Gamma(s)$ gives

$$\Gamma(k+\alpha) = \Gamma(1+\alpha) \prod_{j=1}^{k-1} (j+\alpha) \geq \frac{1}{2} \prod_{j=1}^{k-1} j = \frac{1}{2} (k-1)!,$$

where we used $\Gamma(1+\alpha) \geq \frac{1}{2}$ and $j+\alpha \geq j$ (the product is empty for $k = 1$). Substituting into $g_\alpha(x, y)$:

$$\begin{aligned} g_\alpha(x, y) &\leq e^{-x-y} \left[2x^{\alpha-1} + \sum_{k \geq 1} \frac{2x^{k+\alpha-1} y^k}{(k-1)! k!} \right] \\ &= 2x^{\alpha-1} e^{-x-y} + 2x^\alpha e^{-x} \underbrace{\sum_{k \geq 1} \frac{x^{k-1}}{(k-1)!} \cdot \frac{y^k e^{-y}}{k!}}_{\leq e^x \cdot 1} \leq 2x^{\alpha-1} (e^{-x} + x), \end{aligned}$$

using $y^k e^{-y}/k! \leq \sup_y (y^k e^{-y}/k!) \leq 1$ (term in a Poisson distribution), and $\sum_{k \geq 1} x^{k-1}/(k-1)! = e^x$. Thus

$$g_\alpha(x, y) \leq 2x^{\alpha-1}(e^{-x} + x), \quad (14)$$

and substituting into Theorem 3.6 (note Y_i drops out),

$$h_\alpha(x) \leq (2/\mu)^n \prod_i (x_i/\mu)^{\alpha_i-1} (1 + x_i/\mu). \quad (15)$$

When $\alpha_i > 1$, the factor $(x_i/\mu)^{\alpha_i-1} \rightarrow 0$ as $x_i \rightarrow 0+$, so $h_\alpha(x) \rightarrow 0$.

- (b) Derivative formula. Differentiating the series in Theorem 3.4 term by term (using $\Gamma(k + \alpha) = (k + \alpha - 1)\Gamma(k + \alpha - 1)$; for the $k = 0$ term this reads $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$, which is exactly where $\alpha > 1$ is needed),

$$\partial_x g_\alpha(x, y) = g_{\alpha-1}(x, y) - g_\alpha(x, y), \quad \alpha > 1, \quad (16)$$

termwise differentiation being justified by the Weierstrass M -test: for $k \geq 1$ and $x \in [0, d]$ the x -derivative of the k -th term is bounded by $((k + \alpha - 1)d^{k+\alpha-2} + d^{k+\alpha-1})y^k/(\Gamma(k + \alpha)k!)$, a summable sequence, so the differentiated series converges uniformly on compact subsets of $(0, \infty)$. Inserting into Theorem 3.6: the integrand and its x_i -derivative are bounded by an integrable function of (x, ω) uniformly in x_i on compact subsets of $(0, \infty)$ (use (14) for $g_{\alpha_i-1}(\cdot, Y_i)$ and $g_{\alpha_i}(\cdot, Y_i)$), justifying dominated convergence to differentiate under $\mathbb{E}$, which gives (11).

- (iii) Iterating (11),

$$\frac{\partial^{|J|}}{\partial x_J} h_\alpha = \frac{1}{\mu^{|J|}} \sum_{\delta \in \{0,1\}^J} (-1)^{|J|-|\delta|} h_{\alpha - \sum_{i \in J} \delta_i e_i}, \quad (17)$$

each summand in L_1 by (i) (all $\alpha_i - \delta_i \geq \alpha_i - 1 > 0$). Integrate by parts in each x_j , $j \in J$, in turn. At the step removing ∂_{x_j} , the function being differentiated is a finite combination of densities $h_{\alpha'}$ (via (17)) in which coordinate j is *not* among the lowered ones, so its j -th shape parameter is still $\alpha_j > 1$; by (i) each such $h_{\alpha'}$, and by (17) its x_j -partial, lies in L_1 . Hence, for a.e. fixing of the remaining coordinates, $x_j \mapsto h_{\alpha'}$ is absolutely continuous with itself and its derivative integrable; such a function tends to 0 as $x_j \rightarrow \infty$ (an L_1 function on $(0, \infty)$ with integrable derivative has limit 0 at infinity), and to 0 as $x_j \rightarrow 0+$ by (ii) (since $\alpha'_j = \alpha_j > 1$). Both boundary contributions therefore vanish (the bounded factor $e^{-\sum_i \lambda_i x_i} \in (0, 1]$ does not affect this), and each integration by parts brings down a factor λ_j , giving (12). □

3.5 The family $h_{k,C}$ and its Laplace transform

Definition 3.8. For $C \succ 0$ and $\mu > 0$ with $C - \mu I_n \succ 0$, set $A := (C - \mu I_n)^{1/2}$ (Theorem 1.3, which is well-defined and invertible), so that $C = \mu I_n + AA^\top$. Let $(\xi_j^{(l)})$, $j \leq n$, $l \leq k$, be i.i.d. $\mathcal{N}(0, 1)$, and

$$Y_i := \sum_{l=1}^k \left(\sum_{j=1}^n \frac{1}{\sqrt{2\mu}} \xi_j^{(l)} a_{ij} \right)^2, \quad h_{k,C} := h_{(k/2, \dots, k/2), \mu, Y}. \quad (18)$$

Lemma 3.9. For $\lambda_i \geq 0$, $\int e^{-\sum_i \lambda_i x_i} h_{k,C} dx = |I_n + \Lambda C|^{-k/2}$.

Proof.

1. Scalar Laplace transform. Substitute $u := x/\mu$ and expand the series in Theorem 3.4:

$$\begin{aligned} \int_0^\infty \mu^{-1} e^{-\lambda x} g_\alpha(x/\mu, y) dx &= \int_0^\infty e^{-\lambda \mu u - u - y} \sum_{m \geq 0} \frac{u^{m+\alpha-1}}{\Gamma(m+\alpha)} \frac{y^m}{m!} du \\ &= e^{-y} \sum_{m \geq 0} \frac{y^m}{m!} (1 + \mu\lambda)^{-(m+\alpha)} \\ &= (1 + \mu\lambda)^{-\alpha} \exp\left(-\frac{\mu\lambda}{1+\mu\lambda} y\right), \end{aligned}$$

using $\int_0^\infty u^{m+\alpha-1} e^{-(1+\mu\lambda)u} du = \Gamma(m+\alpha)(1+\mu\lambda)^{-(m+\alpha)}$ (Fubini-term-by-term justified by absolute convergence). So

$$\int_0^\infty \mu^{-1} e^{-\lambda x} g_\alpha(x/\mu, y) dx = (1 + \mu\lambda)^{-\alpha} \exp\left(-\frac{\mu\lambda}{1+\mu\lambda} y\right). \quad (19)$$

2. Product form. Applying (19) coordinatewise with $\alpha_i = k/2$ and Fubini (nonneg integrand),

$$\int e^{-\sum_i \lambda_i x_i} h_{k,C} dx = \prod_i (1 + \mu\lambda_i)^{-k/2} \cdot \mathbb{E} \exp\left(-\sum_i \frac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\right). \quad (20)$$

3. Evaluate the expectation. From (18), set $X^{(l)} := \frac{1}{\sqrt{2\mu}} A \xi^{(l)}$, so $X^{(l)} \sim \mathcal{N}(0, \frac{1}{2\mu} A A^\top)$ i.i.d. across l and $Y_i = \sum_{l=1}^k (X_i^{(l)})^2$. By independence across l and Theorem 3.1 applied to each $X^{(l)}$ with $\tilde{\Lambda} := \text{diag}(\frac{\mu\lambda_i}{1+\mu\lambda_i})$,

$$\mathbb{E} \exp\left(-\sum_i \frac{\mu\lambda_i}{1+\mu\lambda_i} Y_i\right) = \left(\mathbb{E} \exp(-\langle \tilde{\Lambda} X^{(1)}, X^{(1)} \rangle)\right)^k = |I_n + \frac{1}{\mu} \tilde{\Lambda} A A^\top|^{-k/2}.$$

4. Combine. Inserting step 3 into (20) and using $\prod_i (1 + \mu\lambda_i)^{-k/2} = |I_n + \mu\Lambda|^{-k/2}$ (since $I_n + \mu\Lambda$ is diagonal),

$$\int e^{-\sum_i \lambda_i x_i} h_{k,C} dx = (|I_n + \mu\Lambda| \cdot |I_n + \frac{1}{\mu} \tilde{\Lambda} A A^\top|)^{-k/2}.$$

The diagonal identity $\tilde{\Lambda}_i = \mu\lambda_i/(1 + \mu\lambda_i)$ rearranges to $(I_n + \mu\Lambda)\tilde{\Lambda} = \mu\Lambda$. Hence, using the multiplicativity of det,

$$\begin{aligned} |I_n + \mu\Lambda| \cdot |I_n + \frac{1}{\mu} \tilde{\Lambda} A A^\top| &= |(I_n + \mu\Lambda)(I_n + \frac{1}{\mu} \tilde{\Lambda} A A^\top)| \\ &= |I_n + \mu\Lambda + \frac{1}{\mu} (I_n + \mu\Lambda) \tilde{\Lambda} A A^\top| \\ &= |I_n + \mu\Lambda + \Lambda A A^\top| = |I_n + \Lambda(\mu I_n + A A^\top)| = |I_n + \Lambda C|, \end{aligned}$$

giving the claim. □

3.6 Uniqueness of the Laplace transform

Theorem 3.10. If $\varphi_1, \varphi_2 \in L_1((0, \infty)^n)$ satisfy $\int e^{-\sum_i \lambda_i x_i} \varphi_1 dx = \int e^{-\sum_i \lambda_i x_i} \varphi_2 dx$ for all $\lambda \in [0, \infty)^n$, then $\varphi_1 = \varphi_2$ a.e.

Proof. Set $\varphi := \varphi_1 - \varphi_2 \in L_1((0, \infty)^n)$; the hypothesis becomes $\int e^{-\sum_i \lambda_i x_i} \varphi dx = 0$ for all $\lambda \in [0, \infty)^n$.

1. Change of variables. With $y_i := e^{-x_i}$, the map $(0, \infty)^n \rightarrow (0, 1]^n$ has Jacobian $\prod_i y_i^{-1}$, so setting $\tilde{\varphi}(y) := \varphi(-\log y) \prod_i y_i^{-1} \in L_1((0, 1]^n)$ and noting $e^{-\lambda_i x_i} = y_i^{\lambda_i}$,

$$\int_{(0,1]^n} y^\lambda \tilde{\varphi}(y) dy = 0 \quad \text{for all } \lambda \in [0, \infty)^n. \quad (21)$$

2. From polynomials to continuous functions. Restricting (21) to $\lambda \in \mathbb{Z}_{\geq 0}^n$ and taking finite linear combinations, $\int_{[0,1]^n} P(y) \tilde{\varphi}(y) dy = 0$ for every polynomial P (extend $\tilde{\varphi}$ by zero at $y_i = 0$). By Stone–Weierstrass, polynomials are uniformly dense in $C([0,1]^n)$; given $\psi \in C([0,1]^n)$ and $\varepsilon > 0$, pick P with $\|\psi - P\|_\infty < \varepsilon$, so

$$\left| \int \psi \tilde{\varphi} \right| = \left| \int (\psi - P) \tilde{\varphi} \right| \leq \varepsilon \|\tilde{\varphi}\|_1.$$

Sending $\varepsilon \rightarrow 0$: $\int_{[0,1]^n} \psi \tilde{\varphi} dy = 0$ for every $\psi \in C([0,1]^n)$.

3. From continuous functions to $\tilde{\varphi} = 0$ a.e. Let $E := \{\tilde{\varphi} > 0\}$. By inner/outer regularity of Lebesgue measure, there exist compact $K_m \subseteq E$ and open $U_m \supseteq E$ with $|U_m \setminus K_m| \rightarrow 0$; Urysohn gives $\psi_m \in C([0,1]^n)$ with $\mathbf{1}_{K_m} \leq \psi_m \leq \mathbf{1}_{U_m}$. Then $\psi_m \rightarrow \mathbf{1}_E$ a.e. along a subsequence, and $|\psi_m \tilde{\varphi}| \leq |\tilde{\varphi}| \in L_1$, so dominated convergence gives $\int_E \tilde{\varphi} = \lim \int \psi_m \tilde{\varphi} = 0$. Similarly for $\{\tilde{\varphi} < 0\}$, so $\int |\tilde{\varphi}| = 0$ and $\tilde{\varphi} = 0$ a.e., hence $\varphi = 0$ a.e. □

4 Proof of Theorem 2.4

4.1 Reduction to monotonicity

With $s_i := \frac{1}{2}t_i^2$, $K := [0, s_1] \times \cdots \times [0, s_n]$, and $Z(\tau)$ as above, (3) is equivalent to $P(Z(1) \in K) \geq P(Z(0) \in K)$. At $\tau = 0$, $C(0) = \text{diag}(C_{11}, C_{22})$ is block-diagonal, so the two blocks are independent and the RHS factorizes. At $\tau = 1$ we recover C . Hence Theorem 2.4 follows from:

Proposition 4.1. $\tau \mapsto \int_K f(x, \tau) dx$ is non-decreasing on $[0, 1]$.

4.2 Laplace-transform computation of $\partial_\tau f$

Theorem 4.1 amounts to $\int_K \partial_\tau f(x, \tau) dx \geq 0$ for each $\tau \in (0, 1)$ (Theorem 3.3(ii) with $\lambda = 0$ justifies $\partial_\tau \int_K f = \int_K \partial_\tau f$). We obtain this by:

- (a) computing the Laplace transform of $\partial_\tau f$ and matching it with that of a sum of derivatives of h_τ ;
- (b) using Laplace injectivity (Theorem 3.10) to get an a.e. pointwise identity;
- (c) integrating this identity over K .

Step (a). Since $Z_i(\tau) = \frac{1}{2}X_i(\tau)^2$, applying Theorem 3.1 to $X(\tau) \sim \mathcal{N}(0, C(\tau))$ with $\lambda_i/2$ in place of λ_i ,

$$\int_{[0, \infty)^n} e^{-\sum_i \lambda_i x_i} f(x, \tau) dx = \mathbb{E} e^{-\sum_i \lambda_i Z_i(\tau)} = |I_n + \Lambda C(\tau)|^{-1/2}.$$

Differentiating in τ , with Theorem 3.3(ii) (taking $K = [0, \infty)^n$) justifying the interchange:

$$\int_{[0, \infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x, \tau) dx = \partial_\tau |I_n + \Lambda C(\tau)|^{-1/2}. \quad (22)$$

Expand via Theorem 3.2(i) applied to $A = \Lambda C(\tau)$:

$$|I_n + \Lambda C(\tau)| = 1 + \sum_{\emptyset \neq J \subseteq [n]} |(\Lambda C(\tau))_J| = 1 + \sum_{\emptyset \neq J \subseteq [n]} |C(\tau)_J| \prod_{j \in J} \lambda_j, \quad (23)$$

using $|(\Lambda C(\tau))_J| = |\Lambda_J| |C(\tau)_J| = \prod_{j \in J} \lambda_j \cdot |C(\tau)_J|$ since Λ is diagonal.

The block structure of $|C(\tau)_J|$. For $J \subseteq [n]$, set $J_1 := [n_1] \cap J$ and $J_2 := J \setminus [n_1]$. If $J_1 = \emptyset$ or $J_2 = \emptyset$, then $C(\tau)_J = C_J$ is τ -independent, since the off-diagonal blocks of $C(\tau)$ (where the τ appears) are not sampled. Otherwise, writing $C(\tau)_J$ in block form with blocks indexed by J_1, J_2 ,

$$C(\tau)_J = \begin{pmatrix} C_{J_1} & \tau C_{J_1 J_2} \\ \tau C_{J_2 J_1} & C_{J_2} \end{pmatrix},$$

and applying Theorem 3.2(ii),

$$|C(\tau)_J| = |C_{J_1}| |C_{J_2}| \prod_{i=1}^{|J_1|} (1 - \tau^2 \mu_{J_1, J_2}(i)), \quad (24)$$

where the $\mu_{J_1, J_2}(i) \in [0, 1]$ are the eigenvalues of $C_{J_1}^{-1/2} C_{J_1 J_2} C_{J_2}^{-1} C_{J_2 J_1} C_{J_1}^{-1/2}$ (bounded by Theorem 3.2(ii) since $C_J \succ 0$). Differentiating the product in τ ,

$$\partial_\tau |C(\tau)_J| = -|C_{J_1}| |C_{J_2}| \sum_{i=1}^{|J_1|} 2\tau \mu_{J_1, J_2}(i) \prod_{l \neq i} (1 - \tau^2 \mu_{J_1, J_2}(l)) \leq 0 \quad \text{on } [0, 1].$$

Define

$$a_J(\tau) := -\partial_\tau |C(\tau)_J| \geq 0 \quad (\tau \in [0, 1]), \quad (25)$$

with $a_J := 0$ if $J_1 = \emptyset$ or $J_2 = \emptyset$. Then

$$\partial_\tau |I_n + \Lambda C(\tau)| = - \sum_{\emptyset \neq J \subseteq [n]} a_J(\tau) \prod_{j \in J} \lambda_j. \quad (26)$$

Apply the chain rule $\partial_\tau (u^{-1/2}) = -\frac{1}{2} u^{-3/2} \partial_\tau u$ to $u = |I_n + \Lambda C(\tau)|$: (22) becomes

$$\int_{[0, \infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x, \tau) dx = -\frac{1}{2} |I_n + \Lambda C(\tau)|^{-3/2} \partial_\tau |I_n + \Lambda C(\tau)|,$$

and substituting (26) (the two minus signs cancel):

$$\int_{[0, \infty)^n} e^{-\sum_i \lambda_i x_i} \partial_\tau f(x, \tau) dx = \sum_{\emptyset \neq J \subseteq [n]} \frac{1}{2} a_J(\tau) |I_n + \Lambda C(\tau)|^{-3/2} \prod_{j \in J} \lambda_j. \quad (27)$$

4.3 Identification via $h_\tau := h_{3, C(\tau)}$

Here $h_{3, C(\tau)}$ is formed as in Theorem 3.8 with any fixed $\mu \in (0, \lambda_{\min}(C(\tau)))$, which is admissible because $C(\tau) \succ 0$; by Theorem 3.9 its Laplace transform does not depend on μ , so neither does anything below.

By Theorem 3.9 with $k = 3$ and Theorem 3.7(iii) (valid since $\alpha_i = \frac{3}{2} > 1$),

$$\int e^{-\sum_i \lambda_i x_i} \frac{\partial^{|J|}}{\partial x_J} h_\tau dx = \left(\prod_{j \in J} \lambda_j \right) |I_n + \Lambda C(\tau)|^{-3/2}.$$

Comparing with (27): $\partial_\tau f(\cdot, \tau)$ and $\sum_J \frac{1}{2} a_J(\tau) \partial^{|J|} h_\tau / \partial x_J$ are L_1 with the same Laplace transform, so by Theorem 3.10,

$$\partial_\tau f(x, \tau) = \sum_{\emptyset \neq J \subseteq [n]} \frac{1}{2} a_J(\tau) \frac{\partial^{|J|}}{\partial x_J} h_\tau(x) \quad \text{a.e. on } (0, \infty)^n. \quad (28)$$

4.4 Integration over K

Integrate (28) over $K = \prod_j [0, s_j]$. For $J^c := [n] \setminus J$ and $(s_J, x_{J^c})_i = s_i$ ($i \in J$), $= x_i$ ($i \in J^c$), the fundamental theorem of calculus in each x_j , $j \in J$, gives

$$\int_K \frac{\partial^{|J|}}{\partial x_J} h_\tau(x) dx = \int_{\prod_{j \in J^c} [0, s_j]} h_\tau(s_J, x_{J^c}) dx_{J^c} \geq 0, \quad (29)$$

since the lower-endpoint contributions vanish by Theorem 3.7(ii) ($\alpha_j = \frac{3}{2} > 1$), and $h_\tau \geq 0$. Since also $a_J \geq 0$,

$$\int_K \partial_\tau f(x, \tau) dx = \sum_{\emptyset \neq J \subseteq [n]} \frac{1}{2} a_J(\tau) \int_{\prod_{j \in J^c} [0, s_j]} h_\tau(s_J, x_{J^c}) dx_{J^c} \geq 0,$$

which proves Theorem 4.1 and hence Theorem 2.4. $\square$

5 Šidák's inequality

Corollary 5.1 (Šidák). For $X \sim \mathcal{N}(0, C)$ and $s_i \geq 0$,

$$\mathbb{P}(|X_i| \leq s_i, i = 1, \dots, n) \geq \prod_{i=1}^n \mathbb{P}(|X_i| \leq s_i).$$

Proof. If some $s_i = 0$, then either $C_{ii} > 0$, so $\mathbb{P}(|X_i| \leq 0) = 0$ and the right side vanishes, or $C_{ii} = 0$, so $X_i = 0$ a.s. and the i -th constraint holds a.s. and can be dropped from both sides. Hence we may assume $s_i > 0$ for all i , as Theorem 2.4 requires. Induct on n . Case $n = 1$ is trivial. For the step, take $K_1 := \{|x_1| \leq s_1\}$ and $L_1 := \{|x_i| \leq s_i, 2 \leq i \leq n\}$ (both closed symmetric convex). Theorem 2.4 (applied with $n_1 = 1$, $n_2 = n - 1$, and extended to singular C by Theorem 2.5) gives $\mathbb{P}(X \in K_1 \cap L_1) \geq \mathbb{P}(X \in K_1)\mathbb{P}(X \in L_1)$; apply induction to $\mathbb{P}(X \in L_1)$. $\square$

6 Worked examples

We close with three exact demonstrations of the GCI on concrete symmetric convex sets, ordered from a fully rational anchor through the equality boundary to a genuinely correlated case in which strictness is established in closed form. No numerical approximation is used: every probability is given exactly, in terms of e , π , and the error function $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} du$. We use repeatedly that, for $Z \sim \mathcal{N}(0, 1)$ with CDF Φ ,

$$\mathbb{P}(|Z| \leq a) = \Phi(a) - \Phi(-a) = 2\Phi(a) - 1 = \operatorname{erf}(a/\sqrt{2}), \quad (30)$$

the last equality because $\Phi(a) = \frac{1}{2}(1 + \operatorname{erf}(a/\sqrt{2}))$.

6.1 A fully rational anchor: nested disks

We first record the Gaussian measure of a centred disk. For $G \sim \mathcal{N}(0, I_2)$, passing to polar coordinates $x = (r \cos \theta, r \sin \theta)$ with Jacobian r ,

$$\mu(\{\|x\| \leq R\}) = \int_0^{2\pi} \int_0^R \frac{1}{2\pi} e^{-r^2/2} r dr d\theta = \int_0^R r e^{-r^2/2} dr = \left[-e^{-r^2/2} \right]_0^R = 1 - e^{-R^2/2}. \quad (31)$$

(Equivalently $\|G\|^2 \sim \chi_2^2$, an exponential law with mean 2.)

Example 6.1 (Concentric disks). Let $G \sim \mathcal{N}(0, I_2)$ and take the concentric disks $K = \{\|x\| \leq R_1\}$ and $L = \{\|x\| \leq R_2\}$ with

$$R_1^2 = 2 \ln 2, \quad R_2^2 = 2 \ln 4 = 4 \ln 2.$$

By (31),

$$\mu(K) = 1 - e^{-\ln 2} = 1 - \frac{1}{2} = \frac{1}{2}, \quad \mu(L) = 1 - e^{-2 \ln 2} = 1 - \frac{1}{4} = \frac{3}{4}.$$

Since $R_1 < R_2$ we have $K \subseteq L$, so $K \cap L = K$ and $\mu(K \cap L) = \frac{1}{2}$. The GCI (1) reads

$$\frac{1}{2} \geq \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8},$$

a strict inequality with both sides rational.

This is the degenerate *nested* regime: whenever $K \subseteq L$ one has $K \cap L = K$, so the GCI collapses to $\mu(K) \geq \mu(K) \mu(L)$, i.e. $\mu(L) \leq 1$, with equality only in the trivial cases $\mu(L) = 1$ or $\mu(K) = 0$. It serves purely as an exact sanity check; the substance is in Theorem 6.4.

6.2 The equality boundary: independent blocks

Example 6.2 (Slab $\times$ disk on disjoint coordinates). Let $\mu = \mathcal{N}(0, I_3)$ and, with $n_1 = 1$ and $n_2 = 2$, set

$$K = \{x \in \mathbb{R}^3 : |x_1| \leq a\}, \quad L = \{x \in \mathbb{R}^3 : x_2^2 + x_3^2 \leq R^2\},$$

both symmetric and convex. Because the standard normal density on $\mathbb{R}^3$ factorises across the two coordinate blocks,

$$\frac{e^{-\|x\|^2/2}}{(2\pi)^{3/2}} = \underbrace{\frac{e^{-x_1^2/2}}{\sqrt{2\pi}}}_{\text{block } \{1\}} \cdot \underbrace{\frac{e^{-(x_2^2+x_3^2)/2}}{2\pi}}_{\text{block } \{2,3\}},$$

and $K \cap L$ is the product set $\{|x_1| \leq a\} \times \{x_2^2 + x_3^2 \leq R^2\}$, Fubini gives

$$\mu(K \cap L) = \mathbb{P}(|X_1| \leq a) \cdot \mathbb{P}(X_2^2 + X_3^2 \leq R^2) = \mu(K) \mu(L).$$

Evaluating the two factors by (30) and (31),

$$\mu(K \cap L) = \mu(K) \mu(L) = \operatorname{erf}\left(\frac{a}{\sqrt{2}}\right) (1 - e^{-R^2/2}),$$

an *exact equality* in (1).

This is the tight case of the GCI. The covariance coupling the K -block $\{1\}$ to the L -block $\{2, 3\}$ is the zero block, $C_{12} = 0$, so the two blocks are independent and equality follows, as the direct computation confirms. (Theorem 7.1 does not apply as stated, since L is a disk rather than a finite intersection of slabs; its “if” direction rests on the same independence.) Geometrically, the two sets constrain orthogonal subspaces, and orthogonality is precisely where no correlation can be gained.

6.3 A correlated box: strictness in closed form

The final example is the genuinely nontrivial one. A correlated Gaussian box probability has no elementary closed form, but its derivative in the correlation does, and this suffices to prove strict inequality exactly. The tool is the following identity.

Lemma 6.3 (Plackett / Price). Let $\Phi_2(h, k; \rho)$ and $\phi_2(h, k; \rho)$ denote the CDF and density of $\mathcal{N}(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix})$, $|\rho| < 1$. Then

$$\frac{\partial}{\partial \rho} \Phi_2(h, k; \rho) = \phi_2(h, k; \rho). \quad (32)$$

Proof. Write the density as the inverse Fourier transform of the bivariate normal characteristic function,

$$\phi_2(h, k; \rho) = \frac{1}{(2\pi)^2} \iint_{\mathbb{R}^2} e^{-i(sh+tk)} e^{-\frac{1}{2}(s^2+2\rho st+t^2)} ds dt.$$

Differentiating under the integral sign (justified by the Gaussian factor, which dominates all s, t -polynomials uniformly for ρ in compact subsets of $(-1, 1)$) brings down $-st$ from the exponent:

$$\frac{\partial}{\partial \rho} \phi_2 = \frac{1}{(2\pi)^2} \iint (-st) e^{-i(sh+tk)} e^{-\frac{1}{2}(\dots)} ds dt.$$

On the other hand ∂_h and ∂_k bring down $-is$ and $-it$ respectively, so $\partial_h \partial_k \phi_2$ produces the factor $(-is)(-it) = -st$ under the same integral. Hence

$$\frac{\partial \phi_2}{\partial \rho} = \frac{\partial^2 \phi_2}{\partial h \partial k}. \quad (33)$$

Now integrate (33) over the quadrant $(-\infty, h] \times (-\infty, k]$; differentiating Φ_2 in ρ under this (u, v) -integral is again justified by dominated convergence, since $\partial_\rho \phi_2 = \partial_u \partial_v \phi_2$ is a polynomial times a Gaussian density whose covariance has eigenvalues $1 \pm \rho$, bounded above and away from 0 for ρ in compact subsets of $(-1, 1)$, hence dominated by an integrable function of (u, v) . Since ϕ_2 and its first partials decay to 0 at $-\infty$, two applications of the fundamental theorem of calculus give

$$\frac{\partial}{\partial \rho} \Phi_2(h, k; \rho) = \int_{-\infty}^h \int_{-\infty}^k \frac{\partial \phi_2}{\partial \rho} dv du = \int_{-\infty}^h \int_{-\infty}^k \frac{\partial^2 \phi_2}{\partial u \partial v} dv du = \phi_2(h, k; \rho). \quad \square$$

Example 6.4 (Two correlated slabs; Šidák with $n_1 = n_2 = 1$). Let $X \sim \mathcal{N}(0, C_\rho)$ on $\mathbb{R}^2$ with $C_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $\rho \in (-1, 1)$, and take the symmetric slabs

$$K = \{|x_1| \leq a\}, \quad L = \{|x_2| \leq a\}, \quad a > 0,$$

so that $K \cap L = [-a, a]^2$. Let μ be the law of X and put $P(\rho) := \mu(K \cap L) = \mathbf{P}(|X_1| \leq a, |X_2| \leq a)$. Then

$$\mu(K \cap L) - \mu(K) \mu(L) = \int_0^{|\rho|} \frac{e^{-a^2/(1+s)} - e^{-a^2/(1-s)}}{\pi \sqrt{1-s^2}} ds > 0 \quad (\rho \neq 0), \quad (34)$$

so the GCI (1) holds *strictly* for every $\rho \neq 0$, with equality iff $\rho = 0$.

Proof. By inclusion-exclusion for the rectangle $[-a, a]^2$, writing $\Phi_2(h, k) = \mathbf{P}(X_1 \leq h, X_2 \leq k)$,

$$P(\rho) = \Phi_2(a, a) - \Phi_2(-a, a) - \Phi_2(a, -a) + \Phi_2(-a, -a).$$

Differentiating in ρ and applying Theorem 6.3 to each term,

$$P'(\rho) = \phi_2(a, a; \rho) - \phi_2(-a, a; \rho) - \phi_2(a, -a; \rho) + \phi_2(-a, -a; \rho). \quad (35)$$

The exponent of ϕ_2 is $Q(h, k) = \frac{h^2 - 2\rho hk + k^2}{2(1 - \rho^2)}$. At the four corners,

$$Q(\pm a, \pm a) = \frac{2a^2 - 2\rho a^2}{2(1 - \rho^2)} = \frac{a^2(1 - \rho)}{(1 - \rho)(1 + \rho)} = \frac{a^2}{1 + \rho}, \quad Q(\pm a, \mp a) = \frac{2a^2 + 2\rho a^2}{2(1 - \rho^2)} = \frac{a^2}{1 - \rho},$$

(each value occurring twice, by $Q(h, k) = Q(-h, -k)$). Hence the like-sign corners share the value $\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-a^2/(1+\rho)}$ and the opposite-sign corners the value $\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-a^2/(1-\rho)}$, and (35) collapses to

$$P'(\rho) = \frac{1}{\pi \sqrt{1 - \rho^2}} \left(e^{-a^2/(1+\rho)} - e^{-a^2/(1-\rho)} \right). \quad (36)$$

For $\rho \in (0, 1)$ we have $0 < 1 - \rho < 1 + \rho$, so $\frac{a^2}{1+\rho} < \frac{a^2}{1-\rho}$ and therefore $e^{-a^2/(1+\rho)} > e^{-a^2/(1-\rho)}$, giving $P'(\rho) > 0$. Moreover the reflection $(x_1, x_2) \mapsto (x_1, -x_2)$ preserves both Lebesgue measure and the box $[-a, a]^2$ while sending C_ρ to $C_{-\rho}$; hence $P(\rho) = P(-\rho)$, and P is even. Thus P is strictly increasing on $[0, 1)$ and strictly decreasing on $(-1, 0]$, with its unique minimum at $\rho = 0$.

At $\rho = 0$ the coordinates are independent, so by (30)

$$P(0) = \mathbf{P}(|X_1| \leq a) \mathbf{P}(|X_2| \leq a) = \operatorname{erf}\left(\frac{a}{\sqrt{2}}\right)^2 = \mu(K) \mu(L).$$

Integrating (36) from 0 to $\rho > 0$ (and using evenness for $\rho < 0$) gives $P(\rho) - P(0) = \int_0^{|\rho|} P'(s) ds$, which is (34); the integrand is strictly positive on $(0, |\rho|]$, so the gap is strictly positive. $\square$

Remark 6.5 (Exact size of the effect as $a \rightarrow 0$). Expanding the integrand of (34) for small a via $e^{-a^2/(1\pm s)} = 1 - \frac{a^2}{1\pm s} + O(a^4)$ gives

$$e^{-a^2/(1+s)} - e^{-a^2/(1-s)} = a^2 \left(\frac{1}{1-s} - \frac{1}{1+s} \right) + O(a^4) = \frac{2a^2 s}{1-s^2} + O(a^4),$$

so, with the substitution $u = 1 - s^2$ in $\int_0^\rho s(1-s^2)^{-3/2} ds = [(1-s^2)^{-1/2}]_0^\rho = (1-\rho^2)^{-1/2} - 1$,

$$\mu(K \cap L) - \mu(K)\mu(L) = \frac{2a^2}{\pi} \left((1-\rho^2)^{-1/2} - 1 \right) + O(a^4).$$

Since $\text{erf}(a/\sqrt{2}) = \sqrt{2/\pi} a + O(a^3)$ gives $\mu(K)\mu(L) = \frac{2a^2}{\pi} + O(a^4)$, the relative discrepancy tends to an exact constant,

$$\frac{\mu(K \cap L)}{\mu(K)\mu(L)} \xrightarrow{a \rightarrow 0} \frac{1}{\sqrt{1-\rho^2}} > 1 \quad (\rho \neq 0),$$

quantifying exactly how far the inequality is from tight for small boxes.

Remark 6.6 (Geometric reading and the general Šidák inequality). Under the rotation-invariant law $\mathcal{N}(0, I_2)$, two slabs of half-width a whose unit normals u_1, u_2 meet at angle θ define the forms $X_i = \langle u_i, G \rangle \sim \mathcal{N}(0, 1)$ with $\text{cov}(X_1, X_2) = \langle u_1, u_2 \rangle = \cos \theta$. Thus Theorem 6.4 with $\rho = \cos \theta$ says that the rhombic cross-strip $K \cap L$ always carries at least the product measure, strictly so unless $\theta = \frac{\pi}{2}$; perpendicular slabs are independent and recover the equality of Theorem 6.2. Theorem 6.4 is moreover a self-contained, elementary proof of Theorem 5.1 in the case $n = 2$ with equal standardized half-widths and nonsingular C with $C_{ii} > 0$ (the cases $|\rho| = 1$ or $C_{ii} = 0$ being trivial: the two events coincide, or one constraint holds almost surely), independent of the machinery of Section 4; the general n case is the content of Theorem 5.1 (via Theorem 2.4), and was first obtained by Šidák (1967) through a related differentiation in the correlations.

6.4 Statistical applications

The inequality was conjectured and first studied because of the multiple-inference problem below, where it furnishes guarantees that hold *uniformly over the unknown dependence structure*. We give three worked applications: dependence-free simultaneous inference, an exact measure of its conservativeness, and the extension to estimated scale (the multivariate t). Here Φ and Φ^{-1} are the standard normal CDF and quantile.

Example 6.7 (Šidák simultaneous intervals and familywise error). Let $\hat{\theta} \sim \mathcal{N}(\theta, \Sigma)$ in $\mathbb{R}^m$ with known variances $\sigma_i^2 := \Sigma_{ii} > 0$ and *arbitrary* correlation matrix R . Standardizing, $Z := ((\hat{\theta}_i - \theta_i)/\sigma_i)_i \sim \mathcal{N}(0, R)$ with each $Z_i \sim \mathcal{N}(0, 1)$. For $c > 0$ form the rectangle $\mathcal{R}_c := \prod_{i=1}^m [\hat{\theta}_i - c\sigma_i, \hat{\theta}_i + c\sigma_i]$. Then

$$\mathbb{P}(\theta \in \mathcal{R}_c) = \mathbb{P}\left(\bigcap_{i=1}^m \{|Z_i| \leq c\}\right) \geq \prod_{i=1}^m \mathbb{P}(|Z_i| \leq c) = (2\Phi(c) - 1)^m, \quad (37)$$

for every R . Hence the Šidák choice

$$c = c_S := \Phi^{-1}\left(\frac{1+(1-\alpha)^{1/m}}{2}\right)$$

yields simultaneous coverage $\geq 1 - \alpha$; dually, testing $H_{0,i} : \theta_i = 0$ by rejecting when $|\hat{\theta}_i/\sigma_i| > c_S$ controls the familywise error rate (FWER) at level $\leq \alpha$ irrespective of R .

Proof. The sets $K_i = \{z : |z_i| \leq c\}$ are symmetric slabs, and $|\hat{\theta}_i - \theta_i| \leq c\sigma_i \iff |Z_i| \leq c$, so the coverage event is $\bigcap_i \{|Z_i| \leq c\}$ and (37) is exactly Theorem 5.1. At $c = c_S$ one has $2\Phi(c_S) - 1 = (1 - \alpha)^{1/m}$, so the lower bound equals $1 - \alpha$. For the testing statement, under $\theta = 0$,

$$\text{FWER} = \mathbb{P}\left(\bigcup_i \{|Z_i| > c\}\right) = 1 - \mathbb{P}\left(\bigcap_i \{|Z_i| \leq c\}\right) \leq 1 - (2\Phi(c) - 1)^m,$$

which is $\leq \alpha$ at $c = c_S$. For general θ , let $I_0 := \{i : \theta_i = 0\}$ be the set of true nulls; for $i \in I_0$ the test statistic $\hat{\theta}_i/\sigma_i$ equals Z_i , and a familywise error means $|Z_i| > c$ for some $i \in I_0$. Applying Theorem 5.1 to $(Z_i)_{i \in I_0} \sim \mathcal{N}(0, R_{I_0})$ gives $\text{FWER} \leq 1 - (2\Phi(c) - 1)^{|I_0|} \leq 1 - (2\Phi(c) - 1)^m$, which is again $\leq \alpha$ at $c = c_S$ (the FWER being 0 if $I_0 = \emptyset$). $\square$

Remark 6.8 (Uniformly sharper than Bonferroni). The Bonferroni rule uses $c_B := \Phi^{-1}(1 - \frac{\alpha}{2m})$, so $2\Phi(c_B) - 1 = 1 - \frac{\alpha}{m}$. By Bernoulli's inequality $(1 - \alpha)^{1/m} \leq 1 - \frac{\alpha}{m}$ (valid since $0 < \frac{1}{m} \leq 1$, strict for $m \geq 2$),

$$2\Phi(c_S) - 1 = (1 - \alpha)^{1/m} \leq 1 - \frac{\alpha}{m} = 2\Phi(c_B) - 1,$$

whence $c_S \leq c_B$. The Šidák rectangle is therefore contained in the Bonferroni rectangle and gives strictly shorter intervals (equivalently, a uniformly more powerful test) at the same guaranteed level for $m \geq 2$.

Example 6.9 (Exact conservativeness for two correlated tests). Take $m = 2$ with $\text{cov}(Z_1, Z_2) = \rho$ (so the correlation is ρ , the marginals being standard). The true simultaneous coverage at threshold c is $P(\rho)$ from Theorem 6.4 with $a = c$, while the dependence-free guarantee (37) is $P(0) = (2\Phi(c) - 1)^2$. By (34) the excess coverage — equivalently, the amount by which the actual FWER falls below the nominal level — is exactly

$$P(|Z_1| \leq c, |Z_2| \leq c) - (2\Phi(c) - 1)^2 = \int_0^{|\rho|} \frac{e^{-c^2/(1+s)} - e^{-c^2/(1-s)}}{\pi\sqrt{1-s^2}} ds \geq 0.$$

The procedure is thus conservative under any dependence, with slack increasing in $|\rho|$. Two regimes are explicit: as $c \rightarrow 0$ (a very stringent level) the ratio of true to guaranteed coverage tends to $(1 - \rho^2)^{-1/2}$ by Theorem 6.5; and as $\rho \rightarrow 1^-$ (perfectly dependent statistics) the two events coincide, $P(\rho) \rightarrow 2\Phi(c) - 1$ (indeed $(Z_1, Z_2) \Rightarrow (Z_1, Z_1)$ in law as $\rho \rightarrow 1^-$, and the limit event $\{|Z_1| \leq c\}$ has boundary $\{|Z_1| = c\}$ of probability 0), so the excess approaches its supremum (not attained for $\rho \in (-1, 1)$)

$$(2\Phi(c) - 1) - (2\Phi(c) - 1)^2 = 2(2\Phi(c) - 1)(1 - \Phi(c)).$$

This makes precise the familiar fact that Šidák (and a fortiori Bonferroni) corrections are most conservative when the test statistics are strongly correlated.

Example 6.10 (Unknown variance: Šidák for the multivariate t). When the scale is estimated, the standardized statistics follow a multivariate t . Model $X = (Z_1/S, \dots, Z_m/S)$ with $Z \sim \mathcal{N}(0, R)$ (R a correlation matrix) and $S > 0$ independent of Z ; if $\nu S^2 \sim \chi_\nu^2$ then X has the central $t_\nu(R)$ law and each $X_i \sim t_\nu$. Writing F_ν for the t_ν CDF, for any thresholds $c_1, \dots, c_m > 0$,

$$P\left(\bigcap_{i=1}^m \{|X_i| \leq c_i\}\right) \geq \prod_{i=1}^m P(|X_i| \leq c_i) = \prod_{i=1}^m (2F_\nu(c_i) - 1). \quad (38)$$

In particular the Šidák threshold built from t_ν -quantiles controls simultaneous coverage and FWER for correlated t -statistics under arbitrary R .

Proof. Condition on S . For fixed $s > 0$, $\{|X_i| \leq c_i\} = \{|Z_i| \leq sc_i\}$, so the Gaussian inequality Theorem 5.1, applied to the slabs $\{|z_i| \leq sc_i\}$ and using $Z_i \sim \mathcal{N}(0, 1)$, gives

$$P\left(\bigcap_i \{|X_i| \leq c_i\} \mid S = s\right) \geq \prod_{i=1}^m (2\Phi(sc_i) - 1) =: \prod_{i=1}^m g_i(s).$$

Taking expectations over S ,

$$P\left(\bigcap_i \{|X_i| \leq c_i\}\right) = E\left[P\left(\bigcap_i \{|Z_i| \leq Sc_i\} \mid S\right)\right] \geq E\left[\prod_{i=1}^m g_i(S)\right].$$

Each $g_i(s) = 2\Phi(sc_i) - 1$ is nonnegative and nondecreasing on $(0, \infty)$, so the g_i are simultaneously nondecreasing functions of the single variable S . A product of nonnegative nondecreasing functions is again nondecreasing, so Chebyshev's association inequality $E[fg] \geq E[f]E[g]$ (for nondecreasing f, g of one variable), applied iteratively, yields $E[\prod_i g_i(S)] \geq \prod_i E[g_i(S)]$. (Proof of the association inequality for bounded nondecreasing f, g : let S' be an independent copy of S ; since f and g are both nondecreasing, $(f(S) - f(S'))(g(S) - g(S')) \geq 0$, and taking expectations gives $2E[fg] - 2EfEg \geq 0$. Here all functions involved take values in $[0, 1]$, so every expectation is finite.) Finally $E[g_i(S)] = E[2\Phi(Sc_i) - 1] = P(|Z_i| \leq Sc_i) = P(|X_i| \leq c_i) = 2F_\nu(c_i) - 1$. Combining the three displays gives (38). $\square$

Remark 6.11. The three examples isolate distinct virtues of the GCI for inference: Theorem 6.7 gives coverage and error control that are *free of the dependence* R ; Theorem 6.9 measures the resulting conservativeness exactly; and Theorem 6.10 shows the same guarantees survive estimation of the scale, the case of practical relevance for t -statistics. Notably the last step combines the GCI with the classical Chebyshev correlation inequality.

7 Notes

7.1 Historical remarks

The conjecture originates in statistical work on joint coverage of rectangular confidence regions for multivariate normal means: Dunnett–Sobel (1955), Dunn (1958). Das Gupta, Eaton, Olkin, Perlman, Savage, and Sobel (1972) stated it in geometric form. Partial results before Royen:

- Khatri (1967) and Šidák (1967), independently: one set is a single symmetric slab $\{x : |\langle v, x \rangle| \leq t\}$. Iterating it gives our Theorem 5.1.
- Pitt (1977): $d = 2$.
- Schechtman–Schlumprecht–Zinn (1998): both sets are centered ellipsoids.
- Hargé (1999): one set is a centered ellipsoid.
- Borell (1981): a further special class of bodies in $\mathbb{R}^n$.

Royen (2014) proved the full conjecture and, more generally, the analogous inequality for multivariate gamma vectors $\Gamma_n(\alpha, R)$ with $\alpha > 0$ half-integer; the Gaussian case is $\alpha = \frac{1}{2}$. Latała–Matlak (2015) specialised and reorganised the argument for the Gaussian case, which is the version followed here.

7.2 Extension to singular covariance

The main line of the proof already covers singular covariances: the slab vector X of the reduction may have singular covariance, and this is handled by Theorem 2.5; so Theorem 1.1 is proved for every centered Gaussian μ , including $\mu = \mathcal{N}(0, I_r)$. The following alternative reduction shows directly that a singular μ on $\mathbb{R}^n$ can be replaced by the non-singular $\mathcal{N}(0, I_r)$.

Let $C \succcurlyeq 0$ with $\text{rank } C = r \leq n$. By the spectral theorem, $C = O \text{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0) O^\top$ with O orthogonal and $\lambda_i > 0$. Set $B \in M_{n \times r}$ to have the i -th column equal to $\sqrt{\lambda_i} O_{\bullet, i}$; then $BB^\top = C$ and B has full column rank. If $G \sim \mathcal{N}(0, I_r)$, then $BG \sim \mathcal{N}(0, BB^\top) = \mathcal{N}(0, C)$.

For closed symmetric convex $K \subseteq \mathbb{R}^n$:

- $B^{-1}(K) = \{g \in \mathbb{R}^r : Bg \in K\}$ is closed by continuity of $g \mapsto Bg$;
- convex, since $Bg_1, Bg_2 \in K$ implies $B(\alpha g_1 + (1 - \alpha)g_2) = \alpha Bg_1 + (1 - \alpha)Bg_2 \in K$ by convexity of K ;
- symmetric, since $g \in B^{-1}(K) \iff Bg \in K \iff -Bg \in K \iff -g \in B^{-1}(K)$.

Hence, writing $\mu = \mathcal{N}(0, C)$ and $\nu = \mathcal{N}(0, I_r)$,

$$\mu(K \cap L) = \mathbb{P}(BG \in K \cap L) = \nu(B^{-1}(K) \cap B^{-1}(L)) \geq \nu(B^{-1}(K))\nu(B^{-1}(L)) = \mu(K)\mu(L),$$

where the inequality is Theorem 1.1 in dimension r for $\nu = \mathcal{N}(0, I_r)$, which is proved by the reduction to Theorem 2.4 together with Theorem 2.5.

7.3 The choice $k = 3$

Two constraints pin down the shape parameter.

1. Laplace-side matching. Differentiating $|I + \Lambda C(\tau)|^{-1/2}$ in τ yields a linear combination of $|I + \Lambda C(\tau)|^{-3/2} \prod_{j \in J} \lambda_j$ ((27)). To rewrite this as the Laplace transform of a density, we need an auxiliary h_τ whose Laplace transform is $|I + \Lambda C(\tau)|^{-3/2}$, i.e., $h_\tau = h_{k, C(\tau)}$ with $k = 3$ in Theorem 3.9.
2. Integration by parts. Theorem 3.7(ii)–(iii) require $\alpha_i = k/2 > 1$ both for $h_\alpha(x) \rightarrow 0$ as $x_i \rightarrow 0+$ and for validity of the derivative identity $\partial_x g_\alpha = g_{\alpha-1} - g_\alpha$. $k = 3$ gives $\alpha_i = \frac{3}{2} > 1$.

The method applies whenever the Laplace transform of the underlying squared vector has the form $|I + \Lambda C|^{-\beta}$ with $\beta + 1 > 1$, i.e., $\beta > 0$, and an auxiliary law with Laplace transform $|I + \Lambda C|^{-(\beta+1)}$ and the properties of Theorem 3.7 is available; Theorem 3.8 provides one exactly when $k = 2\beta + 2 \in \mathbb{N}$, i.e. for half-integer β . In the Gaussian case $\beta = \frac{1}{2}$. Royen's multivariate gamma result covers those β for which a law with this Laplace transform exists for the given C ; this includes every half-integer $\beta > 0$, the case stated in Section 7.1.

7.4 Strict positivity of a_J

Under $C \succ 0$, the matrix $C_{J_1}^{-1/2} C_{J_1 J_2} C_{J_2}^{-1} C_{J_2 J_1} C_{J_1}^{-1/2}$ is the matrix of squared canonical correlations between the coordinate blocks J_1 and J_2 . Its eigenvalues $\mu_{J_1, J_2}(i)$ lie in $[0, 1)$: evaluating (24) at $\tau = 1$ gives $|C_J| = |C_{J_1}| |C_{J_2}| \prod_i (1 - \mu_{J_1, J_2}(i))$, and since $C \succ 0$ forces $|C_J|, |C_{J_1}|, |C_{J_2}| > 0$, every factor $1 - \mu_{J_1, J_2}(i)$ is strictly positive, i.e. $\mu_{J_1, J_2}(i) < 1$. Consequently, for any J with $J_1, J_2 \neq \emptyset$ and $C_{J_1 J_2} \neq 0$, some $\mu_{J_1, J_2}(i) > 0$, so $a_J(\tau) > 0$ on $(0, 1]$. At $\tau = 0$, every $a_J(0) = 0$ because each $|C(\tau)_J|$ is even in τ (see (24)).

7.5 Equality in the GCI

Proposition 7.1. Let $X \sim \mathcal{N}(0, C)$, $C \succ 0$, $n = n_1 + n_2$, $t_1, \dots, t_n > 0$, and let $K := \{x \in \mathbb{R}^n : |x_i| \leq t_i, i \leq n_1\}$ and $L := \{x \in \mathbb{R}^n : |x_i| \leq t_i, n_1 < i \leq n\}$ be the two sets of Theorem 2.4. Then equality $\mathbb{P}(X \in K \cap L) = \mathbb{P}(X \in K)\mathbb{P}(X \in L)$ holds iff $C_{12} = 0$.

Proof. If $C_{12} = 0$, then $C(\tau) = \text{diag}(C_{11}, C_{22})$ for all τ , so the two coordinate blocks of X are independent and both sides of (3) are equal.

Conversely, suppose equality. From the fundamental theorem of calculus and (28):

$$\mathbb{P}(X \in K \cap L) - \mathbb{P}(X \in K)\mathbb{P}(X \in L) = \int_0^1 \sum_J \frac{1}{2} a_J(\tau) \int_{\prod_{J^c} [0, s_j]} h_\tau(s_J, x_{J^c}) dx_{J^c} d\tau.$$

Each a_J is polynomial in τ (since $|C(\tau)_J|$ is polynomial in the entries, which are affine in τ), hence continuous on $[0, 1]$ with $a_J \geq 0$. The inner integral is > 0 for every $\tau \in [0, 1]$: $g_{3/2}(\cdot, y) > 0$ on $(0, \infty)$ for each $y \geq 0$, because every term in the series of Theorem 3.4 is nonnegative and the $k = 0$ term equals $x^{1/2} e^{-x-y} / \Gamma(3/2) > 0$; taking expectation over Y preserves strict positivity, so $h_\tau > 0$ on $(0, \infty)^n$ and the integral over any product $\prod_{J^c} [0, s_j]$ of positive lengths is > 0 .

Equality thus forces $a_J(\tau) \equiv 0$ on $[0, 1]$ (by continuity) for every J with $J_1, J_2 \neq \emptyset$. By Section 7.4, this means $C_{J_1 J_2} = 0$ for every such J . Taking $J = \{i, j\}$ with $i \leq n_1 < j \leq n$ yields $C_{ij} = 0$, so $C_{12} = 0$. $\square$

7.6 Verification of (29) for $n = 2$

For $J = \{1\}$:

$$\int_0^{s_1} \int_0^{s_2} \partial_{x_1} h_\tau dx_1 dx_2 = \int_0^{s_2} (h_\tau(s_1, x_2) - h_\tau(0+, x_2)) dx_2 = \int_0^{s_2} h_\tau(s_1, x_2) dx_2.$$

For $J = \{1, 2\}$: $\int_0^{s_1} \int_0^{s_2} \partial_{x_1 x_2}^2 h_\tau = h_\tau(s_1, s_2) - h_\tau(s_1, 0+) - h_\tau(0+, s_2) + h_\tau(0+, 0+) = h_\tau(s_1, s_2)$. Both use $h_\tau(\cdot) \rightarrow 0$ at $0+$ from Theorem 3.7(ii).

7.7 Necessity of symmetry

Without $K = -K$, $L = -L$ the inequality fails. Under $\mu = \mathcal{N}(0, I_2)$ take

$$K := \{x : x_1 \geq 1\}, \quad L := \{x : x_1 \leq -1\}.$$

Both are closed convex, neither is symmetric. Then $\mu(K) = \mu(L) = 1 - \Phi(1) \approx 0.1587$, so $\mu(K)\mu(L) \approx 0.0252$, while $K \cap L = \emptyset$ gives $\mu(K \cap L) = 0 < \mu(K)\mu(L)$.

7.8 Šidák correction for multiple testing

The use of Theorem 5.1 to obtain dependence-free simultaneous confidence intervals and familywise error control, its comparison with the Bonferroni correction, the exact conservativeness in the bivariate case, and the extension to estimated scale (the multivariate t) are developed in Section 6.4 (Theorems 6.7, 6.9 and 6.10).

7.9 Related correlation inequalities

	Measure	Events/functions	Conclusion
GCI	Centered Gaussian on $\mathbb{R}^n$	Symmetric convex K, L	$\mu(K \cap L) \geq \mu(K)\mu(L)$
FKG (1971)	Log-supermodular	Increasing f, g	$E[fg] \geq Ef \cdot Eg$
Harris (1960)	Product on $\{0, 1\}^n$	Increasing events A, B	$P(A \cap B) \geq P(A)P(B)$

FKG concerns measures on a finite distributive lattice and contains Harris as a special case. The GCI is formally unrelated to FKG: symmetric convex sets in $\mathbb{R}^n$ are not monotone for any coordinate order, and monotone events on a lattice are not symmetric.

7.10 Open questions

1. Quantitative GCI. No explicit nontrivial lower bound on the excess $\mu(K \cap L) - \mu(K)\mu(L)$ is known in the generality of Theorem 1.1. Candidates depending on Gaussian widths, eccentricity, and the spectral gap of C have been proposed but not established.
2. Non-Gaussian extensions. Counterexamples rule out general log-concave measures. The exact class of product-type inequalities valid for symmetric convex sets — beyond the Gaussian and multivariate-gamma families of Royen — is open.

Remark 7.2 (Functional form). The functional form of the GCI is not an open question: for centered Gaussian X in $\mathbb{R}^n$ and even quasi-concave Borel functions $f, g : \mathbb{R}^n \rightarrow [0, \infty)$ (quasi-concave means that every superlevel set $\{f > s\}$, $s \geq 0$, is convex), $E[f(X)g(X)] \geq Ef(X)Eg(X)$ (in $[0, \infty]$), and taking indicators of closed symmetric convex sets recovers (1). Indeed, if the covariance C is singular, write $X \stackrel{d}{=} BG$ with $G \sim \mathcal{N}(0, I_r)$ as in Section 7.2; $f \circ B$ and $g \circ B$ are again even and quasi-concave, so we may assume $C \succ 0$. By Tonelli,

$$E[f(X)g(X)] = \int_0^\infty \int_0^\infty P(f(X) > s, g(X) > t) ds dt,$$

$$Ef(X)Eg(X) = \int_0^\infty \int_0^\infty P(f(X) > s)P(g(X) > t) ds dt.$$

Each $A_s := \{f > s\}$ and $B_t := \{g > t\}$ is symmetric (f, g even) and convex (f, g quasi-concave). The boundary of a convex set has Lebesgue measure zero and $\mathcal{N}(0, C)$ has a density, so replacing A_s, B_t by their closures changes none of $P(X \in A_s)$, $P(X \in B_t)$, $P(X \in A_s \cap B_t)$ (note $\overline{A_s} \cap \overline{B_t} \setminus (A_s \cap B_t) \subseteq \partial A_s \cup \partial B_t$). Theorem 1.1 applied to the closures gives $P(f(X) > s, g(X) > t) \geq P(f(X) > s)P(g(X) > t)$ for all s, t , and integrating proves the claim.

References

- [1] T. Royen, “A simple proof of the Gaussian correlation conjecture extended to multivariate gamma distributions,” *Far East J. Theor. Stat.* **48** (2014), 139–145. [arXiv:1408.1028](#).
- [2] R. Łatała and D. Matlak, “Royen’s proof of the Gaussian correlation inequality,” in *Geometric Aspects of Functional Analysis*, Lecture Notes in Math. **2169**, Springer, 2017, pp. 265–275. [arXiv:1512.08776](#) (preprint 2015, the year used in the text).
- [3] S. Das Gupta, M. L. Eaton, I. Olkin, M. Perlman, L. J. Savage, and M. Sobel, “Inequalities on the probability content of convex regions for elliptically contoured distributions,” *Proc. Sixth Berkeley Symp.*, 1972.
- [4] L. D. Pitt, “A Gaussian correlation inequality for symmetric convex sets,” *Ann. Probab.* **5** (1977), 470–474.

- [5] C. G. Khatri, “On certain inequalities for normal distributions,” *Ann. Math. Statist.* **38** (1967), 1853–1867.
- [6] Z. Šidák, “Rectangular confidence regions for the means of multivariate normal distributions,” *J. Amer. Statist. Assoc.* **62** (1967), 626–633.
- [7] G. Schechtman, T. Schlumprecht, and J. Zinn, “On the Gaussian measure of the intersection,” *Ann. Probab.* **26** (1998), 346–357.
- [8] G. Hargé, “A particular case of correlation inequality for the Gaussian measure,” *Ann. Probab.* **27** (1999), 1939–1951.
- [9] C. Borell, “A Gaussian correlation inequality for certain bodies in $\mathbb{R}^n$,” *Math. Ann.* **256** (1981), 569–573.
- [10] C. W. Dunnett and M. Sobel, “Approximations to the probability integral and certain percentage points of a multivariate analogue of Student’s t -distribution,” *Biometrika* **42** (1955), 258–267.
- [11] O. J. Dunn, “Estimation of the means of dependent variables,” *Ann. Math. Statist.* **29** (1958), 1095–1111.
- [12] C. M. Fortuin, P. W. Kasteleyn, and J. Ginibre, “Correlation inequalities on some partially ordered sets,” *Comm. Math. Phys.* **22** (1971), 89–103.
- [13] T. E. Harris, “A lower bound for the critical probability in a certain percolation process,” *Proc. Cambridge Philos. Soc.* **56** (1960), 13–20.